01—09 / COMPLETE REVIEW

Mortgage Delinquency Survival & Competing-Risk Modeling

完整專案回顧 × 統計方法速查 × 英文面試準備｜截至 2026-10-31 完成的分析

BUSINESS QUESTION

When will a mortgage first become 90+ days past due?

根據首次可觀察的 Freddie Mac 月度紀錄起算，以貸款發放時的特徵預測 12／24／36 個月內首次嚴重逾期機率，同時將 payoff 視為 competing event。

50,000	2.84M	49,968	2,927
Sample mortgages	Monthly records	Modeling loans	Ever 90+ DPD events

結論先記住：不是 XGBoost 大勝 Cox

Held-out test • 36-month AUC

0.707 / 0.709

Cox CIF / XGBoost CIF；排序能力幾乎相同。

Held-out test • Top-30% event capture

58.2%

兩個模型的最高風險 30% 皆捕捉到 160 / 275 件事件。

專案的核心價值：從可信的事件時間定義、逐月 ETL/QC、competing-risk CIF，到 discrimination、calibration 及 held-out test，建立完整且能解釋的風險建模流程。XGBoost 的高風險群校準較接近觀察值，但整體勝出證據有限。

這份 PDF 怎麼用

- **忘記專案：**先讀第 2–5 頁（資料與 outcome 定義）。
- **複習統計：**讀第 6–11 頁（AJ、Cox、XGBoost、CIF、test）。
- **準備面試：**直接看最後 3 頁（英文 pitch、常見 Q&A、CV）。

重要標示：「09」代表邏輯上的 Final Test Evaluation 階段；實際程式目前放在 Notebook 08 末尾，並未單獨建立 Notebook 09。整個專案沒有宣稱 external／out-of-time validation。

PROJECT MAP

01–09 分別做了什麼？

先掌握 workflow，面試時再依問題深入。

01–02 • DATA
Ingestion & Audit
Pipe-delimited TXT → Raw Delta；核對 loan ID、schema、月度紀錄。

03–04 • OUTCOME
Time-to-event
首次 90+ DPD、payoff、行政 censoring；時間起點與 tie QC。

05 • FEATURES
Modeling Base
Origination covariates、structural missingness、loan-level split。

06 • NONPARAMETRIC
AJ / KM / FICO
比較 competing risk 下的累積風險，描繪未調整風險差異。

07 • MODELING
Cox / XGBoost
Cause-specific hazard、C-index、PH diagnostics、SHAP。

08–09 • EVALUATION
CIF / Validation / Test
個人化機率、AUC/Brier/calibration、top-30% capture、final test。

同一條資料管線：每個 table 的角色

Raw origination
50,000 loans
→
Raw performance
2,837,053 rows
→
Survival table
49,968 loans

Feature table
`freddie_model_base`
→
Split table
`freddie_model_split`
→
Cox & XGB
Competing-risk CIF

研究問題為何要改用 Freddie Mac？

最初探索 Lending Club（約 226 萬筆 loan snapshots），但 `last_pymnt_d` 不是可靠的首次 default 日期，無法精準定義 survival event time。Freddie Mac 有每月 performance history，因此能由實際的 delinquency status 觀察到 **首次 90+ DPD 月份**。

> **面試亮點：**當資料無法支持統計 estimand，主動改換資料來源，而不是硬把 proxy 當成事件時間。這個選擇比盲目提高模型複雜度重要。

專案產出與目前狀態

Output	狀態 / 注意事項
Databricks Delta ETL、survival / modeling tables	已完成，包含資料品質驗證
AJ/KM、Cox/XGB、CIF、calibration、SHAP	已完成
Frozen-model final held-out test	已完成；程式位於 Notebook 08 末端
Databricks AI/BI Dashboard	正在建立；不宣稱已正式部署

01–02 / ETL

原始資料、Join 與逐月 QC

用 PySpark 處理縱向資料；用 Delta tables 保留每個可重現的階段。

50K	2.84M	0	2017–26
Origination rows • 31 columns	Performance rows • 35 columns	Unmatched loan IDs	Reporting coverage

File layout：最常用的欄位

Source	Raw column	Meaning
Origination	`_c19`	Loan ID
Performance	`_c0` / `_c1`	Loan ID / monthly reporting period (YYYYMM)
Performance	`_c3`	Monthly delinquency code：00 / 01 / 02 / 03+
Performance	`_c8` / `_c9`	Zero-balance code / effective date

輸入檔為 `|` 分隔的無標頭 TXT；時間涵蓋 2017-01 至 2026-06。

最重要的 QC 結果

Check	Result	Why it matters
Orig vs. perf IDs	50,000 unique IDs each；0 unmatched	避免資料 join 漏失
Duplicate loan-month	**0**	一個月不能有重複狀態
Invalid reporting dates	**0**	確認時間軸
Monthly gaps before exit	**0；max gap = 1 month**	沒有缺月導致事件時間漏記
Loan-level survival outcome	**49,968 unique loans**	建模單位一筆貸款一列

> **為何 PySpark？** Monthly performance 是一筆 loan 對多筆月份的 one-to-many 長資料；PySpark 的 window / groupBy / join 與 Delta 很適合建立可重現的 ETL。2.84M rows 不代表 pandas 一定處理不了；重點在 scalable workflow。

Databricks 主要物件

`workspace.default.freddie_orig_raw`
`workspace.default.freddie_perf_raw`
Raw input

`workspace.default.freddie_survival`
`workspace.default.freddie_model_base`
`workspace.default.freddie_model_split`
Derived tables

需要記得：目前分析僅有 2017 sample，不能把 loan-wise holdout 稱為跨年份的 temporal validation。

03–04 / OUTCOME

如何定義 Survival Time？

Endpoint 不只是 yes/no，更重要的是 time origin、event precedence、competing exits。

ESTIMAND • PRECISION MATTERS

Time from first observed reporting month to first observed 90+ DPD

不是 loan origination 後 lifetime first default，也不是 foreclosure 或 charge-off。

Monthly status → survival event

Perf _c3	Interpretation	Rule
00	Current	No target event
01 / 02	30–59 / 60–89 DPD	No 90+ DPD event
03 and above (numeric)	90+ DPD	**First observed month → cause 1**
Non-numeric/special	特殊狀態	不可直接當作 current 或事件

Exit classification：49,968 loans

event_type	Definition	Count
0	Administrative censoring（觀察截止仍未發生事件）	10,830
1	First 90+ DPD	**2,927**
2	Paid off / matured（zero-balance code 01）	36,128
3	Other termination	83

在 08 的 competing-risk prediction 中，cause 2 把 types 2 與 3 合併；這是建模簡化，個別 non-payoff termination 的機制仍值得 sensitivity analysis。

Time origin / censoring audit

- **Time zero**：第一個 Freddie Mac performance report month；與 first payment date 的差距中位數約 −1 個月，600 筆差距 ≥3 個月。
- **First observed 90+ DPD**：用每筆貸款第一個代碼 ≥03 的月份；觀察起始月沒有 90+ DPD。
- **Same-month ties**：10 筆 delinquency 與 termination 同月，主分析暫以 90+ DPD 優先。
- **Zero-time exits**：排除 32 筆沒有正追蹤時間的非事件貸款。
- **Post-cutoff termination**：3 筆日期為 2026-07，改為截至 2026-06 的行政 censoring。

> **高頻面試陷阱**：2,927 / 49,968 = 5.86% 是「整段觀察期出現過事件的 crude fraction」，不是 36-month CIF。也不能將 payoff 當成「未來仍可能發生目標事件但追蹤中斷」的普通 censoring 去解讀真實累積機率。

05 / MODEL DATA

Feature Engineering 與 Data Split

只使用預測時點已可取得的特徵；training-only preprocessing，避免資料洩漏。

主要特徵與轉換

Feature	Construction / interpretation	Key observation
FICO	`fico_per_20 = fico / 20`	mean ≈744.7；8 missing (0.016%)
Original LTV	`ltv_per_10 = orig_ltv / 10`	mean ≈75%；原始值上限 194%
Interest rate	原始利率（百分點尺度）	mean ≈4.19%
Original UPB	`log_orig_upb`	mean balance ≈ $230,091
Num. borrowers	數值特徵	相同的六特徵用於 Cox / XGB
Relief refinance	`Y → 1`；otherwise 0	1,804 loans flagged Y

非常重要：DTI missingness 是 structural

1,806
Missing original DTI • 3.614%

1,804 / 1,804
所有 Relief Refinance = Y 都缺 DTI

DTI 的缺失不是近似 random 的缺失：1,804 筆 Relief Y 全部缺 DTI，Relief N 只有 2 筆缺失。若把 DTI 全部 median-impute，會掩蓋產品計畫與缺失機制。**最終六特徵 Cox/XGB baseline 不納入 DTI，保留 Relief flag。**

另外，LTV >105 的 90 筆全部屬於 Relief loans。這些是需要理解的產品結構，不應草率視為輸入錯誤。

固定的 loan-level 70 / 15 / 15 split

Split	Loans	Ever 90+ DPD events	Use
Train	34,975	2,048	Fit models / training baseline hazards
Validation	7,495	439	Early stopping / model development
Test	7,498	440	一次性 final held-out evaluation

分組是依 outcome type 分層、固定 hash 的 loan-wise split；不是按月份列切分，因此同一貸款不會跨集合。表內 events 是整段 follow-up，並非 36m events（validation 253；test 275）。FICO 缺失用 **training median** 填補，validation/test 不重新估計補值。

Leakage rule：只用 origination-time predictors，不把未來每月的 delinquency、balance trajectory、payoff/termination dates 等 outcome information 當作 baseline covariates。

06 / NONPARAMETRIC

AJ CIF vs. 1 − Kaplan–Meier

先回答「真實累積發生率」再做個人化模型；payoff 是主要 competing risk。

為什麼不能把 1 − KM 當作實際 90+ DPD 機率？

Kaplan–Meier 若把 payoff 當一般 censoring，估計的是移除 competing events 後所對應的 net-risk 類 estimand；它不等於在真實世界 payoff 可能先發生時的 cumulative incidence。

$$CIF_1(t) = \Sigma_{u \le t} S_{all}(u-) \cdot dN_1(u) / Y(u)$$

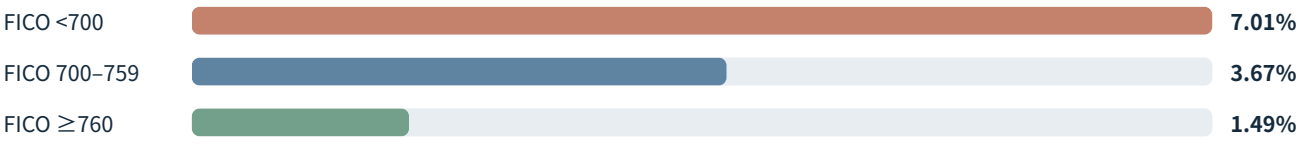
Horizon	AJ CIF • observed world	1 − KM	KM − AJ
12 months	0.42%	0.43%	+0.01 pp
24 months	0.84%	0.89%	+0.05 pp
36 months	**3.32%**	**4.22%**	**+0.90 pp**

ONE NUMBER TO REMEMBER

27.1% higher

36m 時 1−KM 相對 AJ CIF 高約 27.1%（0.90 percentage points），顯示不處理競爭事件會影響風險解讀。這**不是**模型 AUC 改善 27%。

Training-set FICO groups：36-month AJ CIF



Group	Loans	Ever 90+ DPD events	36m AJ CIF
<700	6,493	777	7.0075%
700–759	12,856	850	3.6714%
≥760	15,619	421	1.4918%

> **結果：**FICO <700 與 ≥760 的 36m CIF 相差約 **4.7 倍**。但這是 **unadjusted association**，不能說 FICO 的變動「造成」這個差異。

月度資料的 CIF 曲線約在第 30–40 個 elapsed month 上升較快；橫軸是從 first report 起算的時間，不能直接將這個形狀歸因於特定共同日曆事件。

07 / COX BASELINE

Interpretable Cause-Specific Cox

Cox 用來理解 covariate 與瞬間目標事件 hazard 的關聯，而非直接給 competing-risk absolute probability。

$$\lambda_1(t \mid x) = \lambda_{10}(t) \cdot \exp(x^T\beta) \qquad HR = \exp(\beta)$$

目標事件為 90+ DPD；payoff / other termination 在發生時從 cause-specific risk set 移除。Penalizer = 0.01，六個 origination predictors。HR 為 **adjusted associations**，非 causal effects。

主要 Cox HR（Training Fit）

Covariate / increment	HR	95% CI	Interpretation
FICO / +20 points	**0.815**	0.803–0.828	Hazard 約低 18.5%
LTV / +10 percentage points	1.077	1.051–1.104	Hazard 約高 7.7%
Interest rate / +1 pp	1.587	1.455–1.731	Hazard 約高 58.7%
Log UPB / +1	1.575	1.467–1.692	Hazard 約高 57.5%
Num. borrowers / +1	0.609	0.565–0.657	Hazard 約低 39.1%
Relief flag • Y vs N	0.931	0.778–1.114	CI 包含 1；p≈0.435

C-index：training vs validation

Model	Train C-index	Validation C-index
Cause-specific Cox	0.7270	**0.7306**
XGBoost Cox（下一頁）	0.7423	**0.7369**

Harrell C-index 評估在已觀察到的目標事件與可比較樣本間的 relative-hazard risk ranking；這不是 36m individualized CIF 的 AUC 或 calibration。

Proportional Hazards（PH）Assumption

Scaled Schoenfeld residual test

FICO / +20 points：p ≈ 0.00018（rank transform）；其他五個 covariates 未見 $p < 0.05$。FICO smoothed residual pattern 非完全水平，提示可能有 time-varying effect。

答面試要精準：「XGBoost Cox 可以建模 nonlinear covariate effects，但仍屬 proportional hazards 模型。」所以使用 XGBoost `survival:cox` **沒有**自動解決 FICO 的 PH violation。

07 / XGBOOST + SHAP

Nonlinearity, Ranking & Explainability

與 Cox 使用相同六個 baseline predictors，比較更加彈性的 covariate function。

XGBoost survival:cox 的訓練設定

Objective
`survival:cox` • `cox-nloglik`
tree_method=hist • max_depth=2
learning_rate=0.03

Regularization / early stopping
min_child_weight=20 • reg_lambda=10
n_estimators=800 • early_stopping_rounds=40
Best iteration for 90+ DPD: **437**

Validation horizon-specific AUC on relative-hazard scores

Horizon	Cox score	XGB score	Difference
12m	0.7447	0.7474	+0.0027
24m	0.7412	0.7552	+0.0141
36m	0.7174	**0.7249**	+0.0075

Validation 在 12 / 24 / 36 個月以前行政 censoring 數皆為 0；因此可用 event-by-horizon indicator 做 fixed-horizon AUC，把 payoff 的貸款作非目標事件。12m/24m cases 僅 24 / 64，估計不穩定。

SHAP：哪些特徵推動預測？

Predictor	Mean |SHAP|（約）	Tree splits
FICO / +20	**0.57**	408
Number of borrowers	0.30	131
Log UPB	0.24	421
Original interest rate	0.18	205
LTV / +10	0.14	221
Relief refinance flag	≈0	5

常考：Split count 不是 SHAP importance。Log UPB 用於 421 次 tree splits，比 FICO 的 408 次還多，但 FICO 對模型 raw log-hazard prediction 的平均絕對貢獻最大。

SHAP sanity check： `max |sum(SHAP) − raw_margin| = 2.62 × 10⁻⁶`，符合浮點精度。這驗證加總一致性，不等於模型的機率校準或因果解釋。

不要說：「FICO 在 750 分有真正的生物/金融 threshold」或「Relief 完全沒被模型使用」；tree model 的階梯來自 split，且 Relief 仍出現 5 次分裂。

08 / COMPETING-RISK MODEL

Relative Hazard → Personalized CIF

最重要的 modeling upgrade：不再只有 risk score，而是實際 36-month serious delinquency probability。

Cause 1 hazard
First 90+ DPD

+

Cause 2 hazard
Payoff / termination

→

Individual $CIF_1(t \mid X)$
12 / 24 / 36 months

兩個 cause-specific models 各做兩種演算法

Model family	Cause 1	Cause 2
Cause-specific Cox	90+ DPD hazard	Termination hazard
XGBoost Cox	90+ DPD hazard；best iteration 437	Termination hazard；best iteration 387

$$F_1(t \mid x) = \int_0^t S(u- \mid x)\, d\Lambda_1(u \mid x)$$

$S(t \mid x)$ depends on both cause-specific cumulative hazards Λ_1 and Λ_2

怎麼把 XGB Cox 轉成 CIF？

1. 對每個 cause 分別訓練 XGB Cox，取得各筆貸款的 predicted relative hazard。
2. 只用 **training data** 的 event counts 與 risk sets，計算各 cause 的 **Breslow baseline cumulative-hazard increments**。
3. 對個體 i，每月 hazard increment = baseline increment × individual relative hazard。
4. 結合兩個 cause hazard；依月更新存活機率 S 與 cause 1 CIF，得到 12/24/36m predictions。

> **Piecewise-constant monthly approximation：**每月令 $\Delta H = \Delta H_1 + \Delta H_2$，當月退出機率 = $1-\exp(-\Delta H)$，cause 1 所占比例 ≈ $\Delta H_1/\Delta H$。持續累加 S × exit_probability × cause_fraction；因此 CIF 隨時間非遞減，且在 [0,1]。

為什麼不能把 Cox survival 直接做 1−S？

單一 cause 1 Cox 的 $1-S_1$ 會把其他原因先退出的可能性排除。實際 90+ DPD cumulative incidence 要考慮 payoff 會消耗 risk set；必須合併所有 competing causes 的 total survival。

模型與方法的角色分工

Cause-specific Cox

HR 易解釋；適合 benchmark，並能計算 personalized CIF。

XGBoost Cox CIF

納入 nonlinear covariate effects；仍有 PH 結構、Breslow ties 與月內近似限制。

> **方法界線：**對 other termination (type 3) 一併視為 competing event 是分析假設；像 loan sale／transfer 有可能更接近失訪，其統計分類應做敏感性分析。

08 / VALIDATION

Discrimination ≠ Calibration

Model development set：n = 7,495；36-month 90+ DPD cases = 253。

Validation CIF results

Horizon	Observed	Cox predicted	XGB predicted	Cox / XGB AUC
12m	0.3202%	0.4188%	0.4191%	0.7455 / 0.7482
24m	0.8539%	0.8366%	0.8370%	0.7434 / 0.7573
36m	**3.3756%**	**3.3097%**	**3.3126%**	**0.7195 / 0.7272**

0.03186
Cox CIF 36m Brier score

≈0.0317
XGBoost CIF 36m Brier score

兩種模型的 **overall calibration-in-the-large** 很接近：mean predicted risk 約 3.31%，observed incidence 3.38%。但 overall mean 接近不代表每個 risk decile 都校準良好。

Top 30% high-risk loans（Validation）

Metric	Cox CIF	XGB CIF
Loans	2,249	2,249
Observed events	161	162
Predicted 36m risk	6.091%	6.742%
Observed 36m risk	7.159%	7.203%
Underprediction (O − P)	**1.067 pp**	**0.461 pp**
Event capture	63.636%	64.032%
Risk lift	2.12×	2.13×

Bootstrap：差異有多穩定？

Validation metric	Point	Bootstrap 95% CI
AUC difference (XGB − Cox)	+0.00776	[−0.00207, 0.01690]
Brier reduction (Cox − XGB)	+0.00013	[0.00000, 0.00027]
Cox top-30% underprediction	+1.067 pp	[0.060, 2.224] pp

統計解讀：AUC difference CI 包含 0，不能斷言 XGB 優於 Cox。Brier CI 下限顯示 0.00000（四捨五入），改善很小。Bootstrap 固定模型、validation 曾用於 early stopping；top-30% after seeing plots 具有探索性，不能當確認性檢定。

業務意義：Validation 最高風險 30% 捕捉約 64% 的 36m 事件；但改善 ranking 與把 absolute probability 估準是兩個不同目標。

09 / FINAL TEST (RUN IN 08)

Held-out Test：最後的真實結果

n = 7,498；2017 sample loan-level holdout；固定模型之後評估，未重新調整 hyperparameters。

Final Test AUC：Validation 的 XGB 優勢未穩定延續

Horizon	Cases	Cox CIF AUC	XGB CIF AUC
12m	23	**0.7913**	0.7690
24m	60	**0.7954**	0.7851
36m	**275**	0.7074	**0.7094**

3.6676%
Observed 36m risk • test

≈3.29%
Predicted mean 36m risk • both models

36m Absolute Risk / Brier / Calibration

Metric	Cox CIF	XGB CIF
AUC	0.7074	0.7094
Brier	0.0346	0.0346
Observed incidence	3.6676%	3.6676%
Mean predicted incidence	3.2987%	3.2898%
Prediction − observation	−0.3690 pp	−0.3778 pp

Top 30% group：兩個模型都抓到 160 / 275 cases

Metric	Cox CIF	XGB CIF
Selected mortgages	2,250	2,250
Captured events	**160 (58.18%)**	**160 (58.18%)**
Observed risk	7.111%	7.111%
Predicted risk	6.058%	6.676%
High-risk underprediction	**1.053 pp**	**0.435 pp**

Final interpretation：Cox/XGB 在 36m test discrimination 幾乎相同；XGB 在各自 top-30% cohort 的平均風險較接近觀察結果，但並未捕捉到更多事件。模型整體在 test 低估 36m 風險約 0.37–0.38 pp。

不要過度承諾：12m / 24m cases 只有 23 / 60。這是同一 2017 sample 的 internal test，不是 out-of-time 或外部驗證；test 不能拿來繼續調參。即使 test AUC 0.709，亦不足以宣稱模型可直接用於真實借貸決策。

SYNTHESIS / INTERPRETATION

面試時最值得強調的 5 個發現

用 business value + statistical reasoning 串成故事，不要只背模型指標。

FINDING 1
Data quality first
有可靠 monthly history 才能建立可信 first-event time。

FINDING 2
Payoff matters
36m 1−KM 比 AJ CIF 高 0.90 pp。

FINDING 3
FICO stratifies risk
Low vs high FICO 的 36m AJ CIF 相差約 4.7×。

FINDING 4
ML gains are modest
Held-out 36m AUC：0.707 vs 0.709。

FINDING 5
Calibration matters
XGB top-30% risk 更接近觀察，但 test 整體低估。

BUSINESS IMPACT
Prioritize monitoring
Top-risk 30% 捕捉 test 58.2% 的 36m 事件。

Discrimination / calibration / risk concentration 是三件事

Dimension	Question	Relevant numbers
Discrimination	誰比較高風險？	C-index；36m AUC 0.707 / 0.709
Calibration	預測的機率可信嗎？	Mean predicted vs observed；risk deciles；Brier
Risk concentration	優先管 30% 能抓到多少事件？	160 / 275 = 58.2%

研究限制：面試要主動說出來

- **Time origin / selection：**從 first observed report 起算，而不是 origination；acquisition 前歷史未完整觀察，現有 estimand 只適用可觀察 portfolio entry 後。
- **Termination typing：**除 payoff 外，sale/transfer 等其它終止可能相當於資料中斷；合併為 competing event 是敏感性分析的候選。
- **Monthly granularity：**同月事件與終止無法排序；10 筆 ties 目前 event 優先，並使用 Breslow/monthly approximation。
- **PH assumption：**FICO 的 Schoenfeld test p≈0.00018；XGBoost Cox 同樣沒有 explicit time-varying effects。
- **Rare events / validation：**12m/24m cases 少；validation 用於 early stopping，test 為同一 cohort 內部 holdout。
- **Transportability / fairness：**尚未在不同年份、市場環境及不同群體驗證，不應聲稱可投入真實信貸審批。

下一版最值得投資的改進

> 優先順序：① 其他 vintage 做 out-of-time validation；② termination-code sensitivity + same-month tie sensitivity；③ time-varying FICO effect 或離散時間 hazard model；④ calibration uncertainty 與再校準；⑤ dashboard 的可再現模型版本管理。

INTERVIEW / PITCH

英文面試怎麼介紹？

可依面試時間選擇 30 秒或約 2 分鐘；避免直接背出一大串數字。

30-second pitch • Recruiter screening

I built a mortgage delinquency risk modeling project using Freddie Mac loan-level data. I processed about 2.8 million monthly performance records in PySpark and defined a survival outcome based on the first observed 90-plus-days-past-due event. Because many mortgages are paid off before becoming delinquent, I modeled payoff as a competing risk. I compared Cox and XGBoost survival models, evaluated their cumulative-incidence predictions, and tested both on held-out loans. At 36 months, their test AUCs were about 0.71, and the highest-risk 30% of loans captured 58% of the observed delinquency events.

90–120-second pitch • Data Scientist / Risk Modeling

One project I completed focused on predicting serious mortgage delinquency using Freddie Mac data. The business question was not just whether a mortgage would become delinquent, but when it would first reach 90 or more days past due and how likely that was within 12, 24, or 36 months.

I originally explored a different lending dataset, but realized its snapshots did not reliably identify the event time. I switched to Freddie Mac because it provides monthly loan performance histories. Using Databricks and PySpark, I joined 50,000 mortgage records with roughly 2.8 million monthly observations, audited the timeline, and created a one-row-per-loan survival dataset. I defined the time origin as the first observed reporting month and distinguished delinquency, payoff, other termination, and administrative censoring.

A key statistical challenge was competing risks. At 36 months, the Aalen–Johansen cumulative incidence was 3.32%, while one minus Kaplan–Meier was 4.22%, illustrating why payoff could not simply be ignored for absolute-risk interpretation. I then fit cause-specific Cox and XGBoost Cox models using the same six baseline features, and combined the delinquency and termination hazards to predict individual cumulative incidence.

On the held-out test set, the two models performed similarly: the 36-month AUCs were 0.707 and 0.709. Their highest-risk 30% groups captured 58% of delinquency cases. XGBoost had somewhat better calibration in that group, but the overall gains were modest. The biggest takeaway was that accurate event definitions, competing-risk handling, and calibration mattered at least as much as model complexity.

Follow-up transition：引導面試官問你的優勢

偏 Data Engineering
“The most challenging part was validating the monthly event sequence and preventing outcome leakage.”

偏 Statistical Modeling
“The key distinction was cause-specific hazard versus absolute cumulative incidence.”

偏 Machine Learning
“XGBoost improved validation AUC only modestly, so I prioritized test generalization and calibration over tuning more models.”

偏 Business Analytics
“We quantified how much delinquency risk was concentrated in the highest-scoring mortgages.”

INTERVIEW / TECHNICAL FAQ

常見技術追問：定義與方法

英文答案可以直接拿來練習；內容對應本次實際分析。

1. Why survival analysis instead of binary classification?

The timing matters, and mortgages have different follow-up durations. A binary target without a clear horizon can mix observed non-events with incomplete follow-up. Survival analysis starts from a clearly defined time origin and exit time.

2. Why did you change the original dataset?

The first dataset did not reliably identify when the target event first occurred. I moved to monthly Freddie Mac histories so the endpoint would align with the business question rather than relying on a weak proxy.

3. What exactly counts as an event?

The first observed month with a numeric delinquency status of three or higher, representing at least 90 days past due. It is not necessarily the first lifetime delinquency and is not the same as charge-off or foreclosure.

4. Why is the first reporting month time zero?

Freddie Mac may acquire a mortgage after origination, so the earlier history may not be observable. My estimand is risk since first observed reporting, not risk since loan origination.

5. Why Aalen–Johansen instead of 1 − Kaplan–Meier?

Payoff can prevent later first delinquency on the same mortgage. Aalen–Johansen incorporates the competing event when estimating cumulative incidence. At 36 months, it gave 3.32%, versus 4.22% for one minus Kaplan–Meier.

6. What is the difference between a cause-specific hazard and CIF?

A cause-specific hazard is the instantaneous event rate among loans still at risk. CIF is the actual probability of experiencing a particular event by a horizon, accounting for all competing causes.

> **Interview tip**：所有百分比要分清 **percentage points (pp)** 和 **relative percent**。例如 4.22% − 3.32% = 0.90 pp，而相對差為約 27.1%。

INTERVIEW / FINAL CHEAT SHEET

模型比較、限制與履歷敘述

最後一頁：忘記數字時快速回來看這裡。

7. Did XGBoost solve the PH-assumption violation?

No. XGBoost Cox can learn nonlinear covariate effects but still assumes proportional hazards over time. FICO had evidence of a time-varying association, so a non-PH extension would be useful.

8. What did SHAP tell you?

FICO was the most influential feature on the model’ s log-relative-hazard output. I checked that the SHAP contributions reconstructed the raw model margin within 2.6e−6. These associations are not causal.

9. Was XGBoost better than Cox?

Not decisively. The validation AUC difference had a bootstrap interval including zero. On the held-out test set, the 36-month AUCs were almost identical: 0.707 for Cox and 0.709 for XGBoost.

10. Why check calibration after AUC?

AUC measures ranking, not whether predicted probabilities match actual incidence. Both models identified high-risk mortgages, but they also underestimated absolute risk in the test set.

11. What was the most important data-quality finding?

Missing DTI was structural: all 1,804 relief-refinance loans had missing DTI. I avoided treating those values as ordinary random missingness and used a comparable six-feature baseline instead.

12. What would you improve next?

I would validate on a different origination vintage, revisit ambiguous termination codes and same-month ties, and consider time-varying effects or a discrete-time competing-risk model.

CV bullets • Select one

Technical: Built a PySpark survival modeling pipeline using 2.84M monthly records from 50K Freddie Mac mortgages; developed and evaluated cause-specific Cox and XGBoost competing-risk models for 36-month serious delinquency.

Impact-oriented: Developed mortgage delinquency risk models with ~0.71 held-out 36-month AUC; identified 58% of observed events within the highest-risk 30% of mortgages and assessed probability calibration.

> **Don't claim：** “XGBoost significantly outperformed Cox”；“predicts lifetime default from origination”；“SHAP discovered causal FICO effects”；“externally validated”；“production-ready underwriting.”

復盤一句話：Reliable event times + competing-risk CIF + honest held-out calibration > chasing a tiny AUC improvement.